

1. Introduction

Ten states take almost all of India's foreign direct investment. Since the Department for Promotion of Industry and Internal Trade began reporting state-wise FDI equity figures in October 2019, those ten have absorbed close to 97 per cent of the national total (DPIIT, 2024). A natural question is whether fiscal discipline is part of what separates them from the rest, and whether it sorts investment among them.

Two literatures speak to this and rarely cite each other. One asks how state fiscal policy affects growth: Trivedi and Rajmal (2011) and Panda and Sahay (2022) both report that fiscal deficits contract subsequent growth. A debt-sustainability literature (Misra, Gupta and Trivedi, 2023; NIPFP, 2024) asks whether current borrowing paths are solvent, but stops before asking what those paths do to the investment a state subsequently receives. NITI Aayog's Fiscal Health Index (2025) ranks states on fiscal metrics and validates the ranking against no outcome at all. Internationally, Feld et al. (2024) find fiscal decentralisation matters for FDI location, but the sub-national Indian case is untested.

This note reports what a ten-state, five-year panel can and cannot establish about that question, and the answer is more useful for its negative half. States carrying persistently lower public debt do receive more FDI, the association is large, and five estimators drawn from three families that share no distributional assumptions all recover it with the same sign. But the low-debt states in this sample are the western and southern ones and the high-debt states the northern and eastern ones; seventy-two per cent of the cross-state variance in debt lies between those regional groups. Once regional indicators enter, the debt coefficient is indistinguishable from zero. We report that confound as the finding rather than as a caveat, because it bounds what any panel of this shape can identify, and because the design that would settle the question follows directly from it.

2. Data

The panel is balanced: ten states over five fiscal years, 2019–20 to 2023–24, giving fifty state-year cells. The ten are Maharashtra, Karnataka, Gujarat, Tamil Nadu, Haryana, Telangana, Jharkhand, Rajasthan, West Bengal and Uttar Pradesh — the largest FDI recipients among the states the Sixteenth Finance Commission reports on. Delhi ranks fourth on cumulative inflow but is a union territory, absent from the Commission's annexures, so West Bengal enters in its place. Forty-eight of the fifty cells carry a usable FDI observation; the two that do not belong to Uttar Pradesh, whose DPIIT series begins only in FY 2021–22 and whose earlier cells are treated as missing rather than zero.

Four fiscal indicators, each a share of gross state domestic product, measure discipline: gross fiscal deficit, outstanding liabilities (debt), own tax revenue and capital expenditure. All are computed from the Sixteenth Finance Commission's Volume II annexures (Government of India, 2025), which derive from CAG-audited State Finance Accounts. Capital expenditure is not reported directly and is derived from the identity that the gross fiscal deficit equals the revenue deficit plus capital outlay plus net lending, so it includes net lending; it is therefore a noisy proxy for productive spending, and Section 4 declines to interpret its coefficient. FDI is DPIIT's (2024) state-wise equity series and GSDP is MoSPI's (2024) comparable 2011–12 series, from which nominal growth

is computed year on year. Literacy and urbanisation from Census 2011 (Office of the Registrar General and Census Commissioner, India, 2011) enter the pooled specification only.

Table I reports the range check run before any estimation. One feature anticipates everything that follows: debt-to-GSDP spans 17.6 to 42.5 per cent, and that spread is almost entirely cross-sectional. Decomposing each regressor’s sum of squares, 91.5 per cent of the variation in debt lies between states and 8.5 per cent within them, with a mean pairwise rank correlation of 0.91 between any two years. GSDP growth runs the other way entirely: 98.3 per cent of its variation is within states. Two consequences recur below. The within-state debt coefficient is identified off very little variation and should be expected to be fragile, and the pooled growth specification has almost no cross-state variation to explain.

Table I: Descriptive statistics ($N = 50$; FDI, $N = 48$).

Variable	N	Mean	Std. dev.	Min	Max
Fiscal deficit (% GSDP)	50	2.82	1.26	−0.7	5.4
Debt (% GSDP)	50	28.82	6.55	17.6	42.5
Own tax revenue (% GSDP)	50	6.47	0.80	5.2	8.2
Capital expenditure (% GSDP)	50	2.27	1.01	0.1	4.6
FDI (Rs crore)	48	31,194	43,933	44	163,795
GSDP growth (%)	50	10.39	7.35	−4.38	26.76

3. Method

Two models are estimated. Model 1 takes log FDI as the dependent variable, Model 2 nominal GSDP growth, each on the four fiscal ratios and a COVID dummy:

$$\ln(\text{FDI}_{it}) = \alpha_i + \beta_1 \text{fd}_{it} + \beta_2 \text{debt}_{it} + \beta_3 \text{otr}_{it} + \beta_4 \text{capex}_{it} + \beta_5 \text{covid}_t + \varepsilon_{it} \quad (1)$$

Both are run twice: entity fixed effects with Driscoll-Kraay (1998) standard errors, and pooled OLS with the two Census controls. Keeping both separates the within-state from the between-state dimension, and that distinction turns out to carry the argument.

Ten clusters is far too few for the cluster-robust t -distribution to be reliable, so every coefficient the paper relies on is re-estimated by wild cluster bootstrap imposing the null, with Rademacher weights over the ten states (Cameron, Gelbach and Miller, 2008). That bootstrap, not the cluster-robust p -value, is the arbiter throughout. We also report a Mundlak (1978) correlated random effects specification, in which each regressor enters twice — as the state’s five-year mean and as the annual deviation from it — so that the between-state and within-state coefficients are identified jointly and can be tested against each other inside one equation.

Because the panel is short and one of its five years is an extreme common shock, the same prediction problem is run again through LASSO (Tibshirani, 1996), Elastic Net (Zou and Hastie, 2005) and SHAP values (Lundberg and Lee, 2017) from a shallow random forest (Breiman, 2001), under state-blocked cross-validation. These stay credible at forty-eight observations where instrument-heavy panel GMM estimators do not (Roodman, 2009; Mullainathan and Spiess, 2017). If a penalised linear selector, a tree-based explainer and a panel regression all land on the same variable,

the agreement is worth more than any one p -value — subject to a qualification Section 5 makes precise.

One methodological failure is worth reporting because it is easy to repeat. An early machine-learning specification included the Census controls and ranked urbanisation the dominant predictor. Urbanisation is fixed within a state across all five years, so the forest was using it to identify which state an observation came from, re-learning the fixed effect rather than measuring agglomeration. Both controls are excluded from the machine-learning stage for that reason.

4. Results

4.1 Debt and FDI

The fixed-effects specification ($n = 48$, within $R^2 = 0.429$) and the pooled specification ($n = 48$, $R^2 = 0.643$) disagree about the sign of debt, and both are estimated precisely enough to take seriously (Table II). Pooled, debt is negative: a one-percentage-point lower debt-to-GSDP ratio is associated with roughly 19 per cent higher FDI, since $\exp(0.176) - 1 = 0.192$. Within states, it is positive.

Table II: Model 1, $\ln(\text{FDI})$. Fixed effects with Driscoll-Kraay standard errors, and pooled OLS with Census controls.

Variable	FE + Driscoll-Kraay		Pooled OLS + controls	
	Coef.	p	Coef.	p
Fiscal deficit	+0.566	0.000	+0.567	0.265
Debt-to-GSDP	+0.353	0.000	−0.176	0.007
Own tax revenue	+0.107	0.684	+0.325	0.268
Capital expenditure	−0.173	0.571	−0.366	0.008
COVID year	−1.406	0.000	+0.547	0.161
Literacy	—	—	+0.101	0.035
Urbanisation	—	—	+0.005	0.909

Table III re-estimates these under the wild cluster bootstrap, and the picture thins considerably. The pooled debt coefficient falls to $p = 0.057$, short of five per cent. Literacy and capital expenditure do not survive at all, and no claim rests on either. The within-state fiscal deficit coefficient, highly significant under Driscoll-Kraay, collapses to $p = 0.836$; no deficit-FDI claim is made anywhere in this note.

Table III: Cluster-robust and wild cluster bootstrap p -values, every coefficient relied on. The bootstrap imposes the null and uses Rademacher weights over the ten states.

Coefficient	Estimate	Cluster p	Bootstrap p	Sig. at 5%
M1 pooled: debt	−0.176	0.007	0.057	No
M1 pooled: literacy	+0.101	0.035	0.235	No
M1 pooled: capital expenditure	−0.366	0.008	0.095	No
M1 within: debt	+0.353	0.000	0.031	Yes
M1 within: fiscal deficit	+0.566	0.264	0.836	No
M2 within: debt	+1.097	0.001	0.020	Yes
M2 within: own tax revenue	+4.083	0.045	0.028	Yes
M2 within: COVID dummy	−13.783	0.000	0.003	Yes

4.2 Between states, not within them

The sign reversal is not an artefact of comparing two estimators with different assumptions. The Mundlak specification recovers it inside a single equation (Table IV), and the hypothesis that the within-state and between-state debt coefficients are equal is rejected at $p = 0.017$ under the same bootstrap. A between estimator fitted to the ten state means puts the cross-state coefficient at −0.237, so the pooled estimate is if anything conservative.

Table IV: Debt coefficient in a Mundlak specification of Model 1 and in a between estimator on the ten state means. Standard errors clustered by state.

Specification	Coef.	p	What it identifies
Mundlak: within-state deviation	+0.370	0.0002	Movement around a state's own mean
Mundlak: state-mean term	−0.256	0.0001	Differences between state averages
Test of equality	0.626	0.017	The two dimensions differ
Between estimator	−0.237	0.022	Cross-state relation, direct

The two coefficients answer different questions. The pooled one asks whether a state that persistently carries less debt than its peers attracts more investment than they do; it does. The within-state one asks whether a state's own FDI rises when its own debt rises above its usual level. Over this window it does, and not because borrowing attracts investors: deficits and debt rose everywhere at once in FY 2020–21 while FDI fell everywhere, and inflows then recovered while debt stayed at its elevated level. The within-state estimate describes the COVID window rather than a behavioural relationship, and Section 5 shows it does not survive a full set of year effects.

4.3 Machine-learning agreement, and growth

Both machine-learning families rank debt first among the five fiscal predictors by a wide margin (Table V), agreeing in sign with the pooled regression. Elastic Net is a close relative of LASSO rather than an independent test, and we count them as one verdict.

Table V: Machine-learning importance, five fiscal predictors, state-blocked cross-validation. LASSO and Elastic Net coefficients are standardised; SHAP is mean $|\text{SHAP}|$ from a random forest.

Variable	LASSO	Elastic Net	Mean $ \text{SHAP} $
Debt-to-GSDP	−1.376	−1.280	1.029
Fiscal deficit	+0.563	+0.488	0.219
Capital expenditure	−0.553	−0.562	0.136
Own tax revenue	+0.039	+0.098	0.208
COVID year	+0.129	+0.161	0.007

The growth hypothesis fails outright. Under a single COVID dummy, within-state debt (+1.097) and own tax revenue (+4.083) both clear five per cent under the bootstrap. Under a full set of year fixed effects the own tax revenue coefficient vanishes (+0.896, $p = 0.611$) and the debt coefficient reverses sign (−0.689, $p = 0.086$). Nothing in the growth equation is stable across the two time controls, in either timing, and the pooled form has under two per cent of its dependent variable’s variation between states to explain in the first place. We report the growth hypothesis as unsupported by this panel.

5. The regional confound

Group the ten states by region and the debt ordering is largely a regional ordering. Mean debt runs 20.6 per cent of GSDP in the west (Maharashtra, Gujarat), 26.2 in the south (Karnataka, Tamil Nadu, Telangana), 33.1 in the north (Haryana, Rajasthan, Uttar Pradesh) and 34.5 in the east (Jharkhand, West Bengal). Seventy-two per cent of the cross-state variance in debt lies between those four groups rather than within them. Regional indicators alone explain more of log FDI than debt alone does, at $R^2 = 0.60$ against 0.31.

Table VI reports what happens to each of the paper’s estimates when regional indicators are added. The pooled coefficient reverses sign and becomes indistinguishable from zero. The Mundlak between-state term reverses. The rank correlation collapses. LASSO promotes a western-region indicator above debt. Within regions the ordering is not reliable either: West Bengal carries more debt than Jharkhand and receives more investment; Rajasthan carries more than Haryana and receives less.

Table VI: Every estimate this note relies on, with and without regional indicators. Rank correlation is Spearman’s ρ between five-year mean debt and five-year total FDI across the ten states; “with region” removes regional means from both series.

Method	Statistic	Without region	With region
Pooled OLS	Debt coefficient	−0.176	+0.036
Mundlak	Between-state debt	−0.256	+0.151
Rank correlation	Spearman’s ρ	−0.842	−0.285
LASSO	Standardised debt coef.	−1.376	−0.679
Random forest	Mean $ \text{SHAP} $, debt	1.029	1.006

Only the SHAP ranking is unmoved. That is the weakest of the five as a defence, because a regional indicator is constant within a state across all five years — exactly the property that made urbanisation unusable in Section 3 — so a tree can use it to identify states rather than to measure geography. The defence applies to the machine-learning results and not to the pooled regression or the rank correlation, and those are where the cross-state claim actually lives.

Two things follow. First, the convergence of five methods is not independent evidence against this confound: they converge partly by reading the same confounded cross-section. Second, the confound cannot be resolved by any refinement of this sample. With ten states and four regions, regional effects absorb three of the nine between-state degrees of freedom, and debt and region are close to collinear by construction. A reader who prefers the regional reading cannot be refuted from these data.

That is a statement about the design rather than about the question, and the distinction matters because the Commission's annexures already contain the fiscal side of the remedy. FC-16 reports all four indicators for twenty-eight states and DPIIT reports a cumulative inflow for each. Across the ten states studied here, 72 per cent of the cross-sectional variance in debt lies between regions and the mean within-region spread is 6.7 percentage points. Across twenty-six states with both series available, the between-region share falls to 35 per cent and the mean within-region spread more than doubles to 15.1 points, with six regions averaging over four states each rather than four regions averaging two and a half. A cross-section on those twenty-six states points the same way as this panel: debt alone gives -0.299 ($p = 0.004$), and with regional indicators added it holds at -0.143 ($p = 0.056$) instead of collapsing to zero as it does here. We do not report that as a result — the wider set has no GSDP series in our hands and so no market-size control, and a cumulative total over a different window is not the annual outcome modelled here. We report it as evidence that the identifying design exists and is within reach.

6. Conclusion

Foreign direct investment in India is more likely where a state holds its debt down relative to its own economy, but only between states, not within a state from one year to the next. Five estimators from three families agree on the sign, and the between-within split survives inside a single equation. What none of them can do is separate that fiscal signal from a regional one. Ten states across four regions do not contain the variation required, and the honest report of this panel is the identification problem rather than the coefficient.

Two implications follow for practice. For a state finance department, the reading consistent with these estimates is that fiscal position operates as a structural, between-state signal rather than a lever: a state that has held debt below 25 per cent of GSDP for five consecutive years is in a different competitive tier from one that has not, and no single budget year moves that. For an index builder — NITI Aayog's Fiscal Health Index (2025) being the immediate case — the result is a caution. The composite ranking constructed here is outperformed by debt alone on every outcome tested, and a composite validated against no outcome cannot know whether it is measuring anything.

The extension that would settle the question is specific rather than aspirational: a panel wide enough to hold region fixed while debt still varies within it. Twenty-six states clear that bar where

ten do not.

Data and code availability

The complete dataset, analysis code, results and the full-length working paper from which this note is drawn are openly available at <https://github.com/fbivinay/Fiscal-Discipline-and-Investment-Outcomes>. The repository contains the audited source series, the 50-row analysis panel behind every estimate reported here, and every analysis script.

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